ANGUS CHOU

(778) 668-0388 | tsuki9355.jobs@gmail.com | Portfolio | GitHub | LinkedIn

EDUCATION

University of British Columbia | Computer Science (Co-op) **Sept 2025 — Expected Jun 2029**
Dean's List | GPA: 3.8 | Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering

TECHNICAL SKILLS

Languages: Java, TypeScript, SQL, C++, C#
Frameworks & Cloud: Spring Boot, React, JDBC, MySQL, AWS (Elastic Beanstalk, RDS)
Tools & Concepts: Git, REST APIs, Bash, Object-Oriented Design, Test-Driven Development, Agile/SDLC
Testing: JUnit, Mockito, Vitest, React Testing Library

TECHNICAL PROJECTS

Expense Tracker | **Full-Stack Web Application** | *React, TypeScript, Java, Spring Boot, MySQL, AWS* | GitHub

- Built a full-stack expense tracker with a React/TypeScript frontend and layered **Spring Boot REST API** (controller, service, repository), executing SQL queries via JDBC with SUM() and GROUP BY for spending summaries.
- Designed an accessible, responsive UI with labeled form fields, screen-reader error announcements, keyboard focus states, and dark mode to record expenses, browse history, and visualize category spending.
- Wrote **18 automated tests** across the stack using JUnit and Mockito for business rules and SQL, Vitest and React Testing Library for UI behavior.
- Deployed the frontend and API as a single artifact to **AWS Elastic Beanstalk** with Amazon RDS MySQL, using Spring profiles and environment variables for configuration.

Zoo Manager | *Java, Swing, JUnit, JSON* | GitHub

- Built a turn-based zoo simulation game in Java with 25 player actions across Swing and command-line interfaces. Designed a shared object-oriented model with a day-cycle economy, a 16-species unlock system, and a capacity-limited hospital.
- Achieved full test coverage of the game rules with **96 JUnit tests** across 7 test classes, including deterministic tests of random events using an injected random number generator. Kept JSON save files backward-compatible across **3 format versions.**

My Personal Website | *TypeScript, React, Vite, CSS* | **Portfolio**

- Built a responsive portfolio in React 19 and TypeScript with Vite, using a **component-based architecture and data-driven content** for projects and experience.
- Integrated the **YouTube IFrame API** into a custom music player with play/pause, track skipping, and auto-advance, using refs to avoid stale state in the **player's event callbacks**.

EXPERIENCE

UBC Game Development Club | Gameplay Programmer **Sept 2026 — Present**

- Won **1st place** among 10 teams in Battle of Concepts with a boss-rush action game concept selected for development by a 10-person team.
- Took ownership of the core absorb-to-attack combat mechanic, designing a system where players gain offensive abilities by successfully absorbing enemy attacks.
- Collaborated with a multidisciplinary team to align gameplay systems with combat design goals.

AllSet Learning | Math & English Tutor **Oct 2025 — Present**

- Earned management recognition as a **top-performing mathematics tutor** by tutoring students and communicating detailed progress and learning needs.

CERTIFICATIONS

Amazon Software Development Credential | Certificate **Oct 2026**

- Applied OOP, SQL/JDBC, TDD, Spring Boot, REST APIs, and AWS deployment concepts across the software development lifecycle.